

CMPS 297AF – Course Project Assignment

1. Overview

In this project, you will apply the concepts, tools, and databases covered across the semester to create an original, biologically meaningful analysis or resource.

You may choose between two project paths:

Path A: Tool Maker

Design and implement a novel bioinformatics tool, workflow component, or visualization method.

The goal is not to re-implement existing tools (e.g., BLAST, Clustal, DESeq, STAR), but to contribute something conceptually new, integrative, or creatively useful.

Your tool may be:

- A command-line utility
- A Python/R package
- A web-based mini application
- A workflow extension
- A visualization module
- A scoring or ranking method

You may use Python, R, Java, C/C++, or web frameworks.

Examples (inspiration only)

- A GO enrichment summarizer that clusters redundant terms and generates interpretable plots
- A novel alignment scoring scheme capturing conservation patterns overlooked by classical matrices
- A unified multi-database query interface (UniProt + KEGG + Reactome + NCBI)
- A pathway overlay tool that colors KEGG/Reactome maps from expression data input
- A network submodule identifier and extractor with Cytoscape-ready outputs

These are just examples to illustrate scope and creativity, they are not constraints.

Path B: Tool User

Use existing bioinformatics tools to answer a clear biological question through a coherent computational workflow.

Your project must:

- Ask a well-defined biological question
- Use multiple course concepts
- Produce interpretable biological conclusions

You may integrate any topics covered in the course, including:

- Pairwise alignment
- BLAST and homology analysis
- Multiple sequence alignment + phylogenetic inference
- Database exploration (e.g. NCBI, UniProt, KEGG, Reactome)
- Ontology and enrichment analysis
- Network analysis
- RNA-seq or microarray expression analysis
- NGS file formats, QC, and mapping concepts

Examples (inspiration only):

1. Homology → Phylogeny → Functional Insight

- Retrieve homologs using BLAST
- Construct MSA + phylogenetic tree
- Perform GO/KEGG enrichment
- Visualize functional clusters in a network

2. Expression → Enrichment → Pathways

- Download expression dataset
- QC + normalization + differential expression
- Volcano + heatmap
- GO/KEGG analysis and pathway interpretation
- and/or Cytoscape visualization of gene modules

3. Database-driven investigation

- Start with a gene family or disease gene set
- Explore UniProt, KEGG, Reactome
- Relate structure, function, domains, interactions
- Build an integrative figure or network

Projects may be purely computational or dataset-driven. Students may propose their own biological questions.

2. Project Timeline & Deliverables

Milestone 1: Proposal (10%) [Due March 15th]

Length: 1–2 pages

Must include:

- Group members
- Title + chosen path (Tool Maker or Tool User)
- Scientific motivation
- Datasets and tools you expect to use
- Project plan (steps, expected outputs)
- Anticipated challenges

Milestone 2: Progress Report (15%) [Due April 5th]

Length: 4–6 pages

Include:

- Updated plan and refined question/tool
- Preliminary results (screenshots, alignments, plots, partial networks)
- Any obstacles or adjustments
- Clear roadmap to completion

Final Report (50%) [Due April 30th]

Length: 8–12 pages

Structure:

- Abstract
- Introduction & biological context
- Methods (explicit description of tools, data, pipelines)
- Results (e.g. figures required: alignment, plots, networks, pathway maps)
- Discussion
- Conclusion
- References

Final Presentation (25%)

A 15-minute in-class presentation summarizing:

- Problem / motivation
- Workflow or tool overview
- Key results
- Biological meaning
- Limitations & future work